SWITCH AND ROUTER CONFIGURATION BASIC THEORY

1. Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Create VLANs in Cisco Packet Tracer

Follow these steps to create two VLANs, Sales (VLAN 10) and Support (VLAN 20), on a Cisco switch.

Step 1: Add a switch

1. Open Cisco Packet Tracer.
2. Drag a 2960 switch onto the workspace.
3. Click the switch and open the CLI tab.

Step 2: Enter the configuration commands

Type these commands in the switch CLI:

```
enable
configure terminal
vlan 10
name Sales
exit
vlan 20
name Support
exit
end
show vlan brief
```

Step 3: Verify the VLANs

The `show vlan brief` command should display VLANs similar to these:

VLAN ID	VLAN Name	Status
10	Sales	Active
20	Support	Active

2. Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command.

Hint: Use the interface range command to

select multiple ports efficiently.

VLAN Port

VLAN 10 (Sales): Fa0/1, Fa0/2, Fa0/3

VLAN 20 (Support): Fa0/4, Fa0/5

Configuration Commands:

enable

configure terminal

interface range fa0/1-3

switchport mode access

switchport access vlan 10

exit

interface range fa0/4-5

switchport mode access

switchport access vlan 20

exit

end

show vlan brief

3. Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Static Routing Configuration

Topology: Router 1 — Router 2

IP Addressing:

Device	Interface	IP Address
Router 1	G0/0	192.168.1.1/24
Router 1	G0/1	10.0.0.1/30
Router 2	G0/1	10.0.0.2/30
Router 2	G0/0	192.168.2.1/24

Static Routing Configuration

1. Connect Router 1 and Router 2 using an Ethernet cable.
2. Assign IP addresses: Router 1 – 10.0.0.1/30, Router 2 – 10.0.0.2/30.
3. Configure a static route on Router 1:

```
enable

configure terminal

 ip route 192.168.2.0 255.255.255.0 10.0.0.2

end
```

4. Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication.

Hint: Think about router-on-a-stick or Layer 3 switch concepts.

Simulation: Inter-VLAN Communication:-

1. A device in Sales (VLAN 10) tries to ping a device in Support (VLAN 20).
2. The ping fails because VLANs are separate broadcast domains and cannot communicate directly without Layer 3 routing.
3. Solution: Configure Router-on-a-Stick using router subinterfaces and 802.1Q trunking, or use a Layer 3 switch with VLAN interfaces (SVIs) and enable IP routing.
4. After configuration, devices in both VLANs can communicate through the router or Layer 3 switch.